

## 23 Extended Field Configurations

### 23.2 Homotopy Groups

We learned in the previous section to classify field configurations, at which the Hamiltonian or other functionals are finite, in correspondence with the elements of appropriate homotopy groups. But we have not yet explained in what sense the homotopy groups are *groups*, nor have we given any physical significance to the group structure. As we shall see, there is a natural definition of the multiplication rule for elements of homotopy groups, according to which two extended configurations of fields forming a manifold  $\mathcal{M}$  in  $d$  dimensions, that belong to different elements  $c_1$  and  $c_2$  of  $\pi_d(\mathcal{M})$ , can only fuse continuously to form a configuration belonging to the element  $c_1 \cdot c_2$  of  $\pi_d(\mathcal{M})$ .

We will begin by defining the first homotopy group  $\pi_1(\mathcal{M})$  of an arbitrary manifold  $\mathcal{M}$ , also known as the *fundamental group of the manifold*. As we have seen, the existence of a non-trivial  $\pi_1(G/H)$  for some coset space  $G/H$  is the condition for the topological stability of a vortex line in three dimensions (or a monopole in two dimensions). After considering  $\pi_1(\mathcal{M})$ , we will then move on to more general homotopy groups.

A connected manifold  $\mathcal{M}$  is said to be multiply connected if there is some closed curve of points  $p(z)$  on the manifold, parameterized by a single variable  $z$  with  $0 \leq z \leq 1$  and  $p(0) = p(1)$ , which cannot be contracted to a point by a continuous deformation. Since on a connected manifold we can always continuously deform any such closed curve so that any one point on the curve is anywhere we like on the manifold, we may restrict our attention only to curves for which  $p(0) = p(1) = p_0$ , where  $p_0$  is any fixed point of the manifold, known as the *base point*. Two such closed curves  $p_1(z)$  and  $p_2(z)$  are said to be homotopically equivalent if they can be deformed into each other, that is, if there is a continuous function  $p(z, t)$  for  $0 \leq t \leq 1$ , with

$$p(z, 0) = p_1(z), \quad p(z, 1) = p_2(z);$$

$$p(0, t) = p(1, t) = p_0.$$

The relation of homotopic equivalence is an equivalence relation in the sense of being symmetric, reflexive, and transitive, so it divides the space of closed curves on the manifold into equivalence classes: two closed curves are in the same class if and only if they are homotopically equivalent. The set of these equivalence classes is known as the first homotopy group of the manifold,  $\pi_1(\mathcal{M})$ .

To define the multiplication rule for  $\pi_1(\mathcal{M})$ , choose a standard curve  $p[z; c]$  that starts and ends at the base point  $p_0$  for each equivalence class  $c$  in  $\pi_1(\mathcal{M})$ . For any two equivalence classes  $c_1$  and  $c_2$ , define the ‘product’  $c_1 \cdot c_2$  as the equivalence class containing the curve  $p[z, c_1, c_2]$  that starts at  $p_0$ , follows  $p[z, c_1]$  back to  $p_0$ , and then follows  $p[z, c_2]$  back again to  $p_0$ . Formally, we take

$$p[z, c_1, c_2] \equiv \begin{cases} p[2z, c_1], & 0 \leq z \leq \frac{1}{2}, \\ p[2z - 1, c_2], & \frac{1}{2} \leq z \leq 1. \end{cases}$$

We must now show that multiplication defined in this way satisfies the conditions for a group. First, let us check that this multiplication is associative. For this purpose, note that  $(c_1 \cdot c_2) \cdot c_3$  is the equivalence class containing a curve  $p[z, c_1 \cdot c_2, c_3]$  that goes along the standard curve  $p[z, c_1 \cdot c_2]$  from the base point and back again, and then along the standard curve  $p[z, c_3]$  from the base point and back again, while  $c_1 \cdot (c_2 \cdot c_3)$  is the equivalence class containing a curve  $p[z, c_1, c_2 \cdot c_3]$  that goes along the standard curve  $p[z, c_1]$  from the base point and back again, and then along the standard curve  $p[z, c_2 \cdot c_3]$  from the base point and back again. By definition, the curve  $p[z, c_1 \cdot c_2]$  may be deformed into a curve that goes along  $p[z, c_1]$  from the base point and back again and then along  $p[z, c_2]$  from the base point and back again, while  $p[z, c_2 \cdot c_3]$  may be deformed into a curve that goes along  $p[z, c_2]$  from the base point and back again and then along  $p[z, c_3]$  from the base point and back again. Hence both curves  $p[z, c_1 \cdot c_2, c_3]$  and  $p[z, c_1, c_2 \cdot c_3]$  may be deformed into the curve that goes along  $p[z, c_1]$  from the base point and back again, then along  $p[z, c_2]$  from the base point and back again, and finally along  $p[z, c_3]$  from the base point and back again, and hence they may be deformed into each other, showing that

$$(c_1 \cdot c_2) \cdot c_3 = c_1 \cdot (c_2 \cdot c_3).$$

The unit element  $e$  of  $\pi_1(\mathcal{M})$  is defined as the equivalence class containing the curve  $p[z, e] = p_0$  that stays at the base point. To check that  $e \cdot c = c$ , note that

$$p[z, e, c] = \begin{cases} p_0, & 0 \leq z \leq \frac{1}{2}, \\ p[2z - 1, c], & \frac{1}{2} \leq z \leq 1. \end{cases}$$

But this curve can be continuously deformed into  $p[z, c]$  by taking

$$p(z, t) = \begin{cases} p_0, & 0 \leq z \leq t/2, \\ p[(2z - t)/(2 - t), c], & t/2 \leq z \leq 1, \end{cases}$$

which is the same as  $p[z, e, c]$  for  $t = 1$  and the same as  $p[z, c]$  for  $t = 0$ . The product  $e \cdot c$  is the equivalence class containing  $p[z, e, c]$ , which we now see is the same as the equivalence class containing  $p[z, c]$ , which is just  $c$ . The proof that  $c \cdot e = c$  is similar.

The ‘inverse’  $c^{-1}$  of the equivalence class  $c$  is the equivalence class that contains a curve  $p^{-1}[z, c]$  which goes around the same path as the standard curve  $p[z, c]$  but in the opposite direction; that is,

$$p^{-1}[z, c] = p[1 - z, c].$$

This is not necessarily the same as the ‘standard’ curve  $p[z, c^{-1}]$ , but by the definition of the equivalence class  $c^{-1}$ , the two curves may be deformed into each other. To check that  $c^{-1} \cdot c = e$ , note that by deforming  $p[z, c^{-1}]$  into  $p^{-1}[z, c]$ , the curve  $p[z, c^{-1}, c]$  may be deformed into

$$p[z, c^{-1}, c] \longrightarrow \begin{cases} p[1 - 2z, c], & 0 \leq z \leq \frac{1}{2}, \\ p[2z - 1, c], & \frac{1}{2} \leq z \leq 1. \end{cases}$$

But this curve can be continuously deformed into  $p[z, e] = p_0$  by taking

$$p(z, t) = \begin{cases} p[1 - 2tz, c], & 0 \leq z \leq \frac{1}{2}, \\ p[2tz + 1 - 2t, c], & \frac{1}{2} \leq z \leq 1, \end{cases}$$

which is the same as  $p[z, c^{-1}, c]$  for  $t = 1$  and the same as  $p[z, e]$  for  $t = 0$ . The product  $c^{-1} \cdot c$  is the equivalence class containing  $p[z, c^{-1}, c]$ , which we now see is the same as the equivalence class containing  $p[z, e]$ , which is just  $e$ . The proof that  $c \cdot c^{-1} = e$  is similar. The existence of a unit element and inverses shows that these equivalence classes form a group.

The classic example of a manifold  $\mathcal{M}$  with a non-trivial first homotopy group is the circle itself,  $\mathcal{M} = S_1$ . This may be parameterized by an angle  $\theta$  with  $\theta = 0$  and  $\theta = 2\pi n$  (with  $n$  any positive or negative integer) identified as the same point. The homotopy groups consist of classes of functions  $\theta(z)$  for  $0 \leq z \leq 1$  that begin at some base point,  $\theta(0) = \theta_0$ , and end at the same base point,  $\theta(1) = \theta_0 + 2\pi n$ . Two such functions may be continuously deformed into each other if and only if they have the same value of  $n$ , so  $\pi_1(S_1)$  consists of a denumerably infinite number of classes  $c_n$  labelled by the positive or negative integer  $n$ . Furthermore, the ‘product’ of two classes  $c_n$  and  $c_m$  consists of curves that go  $n$  times around the circle from the base point and back again, and then  $m$  times around the circle from the base point and back again, so here multiplication is just addition:

$$c_n \cdot c_m = c_{n+m}. \quad (23.2.1)$$

and hence

$$\pi_1(S_1) = \mathbb{Z}, \quad (23.2.2)$$

where  $\mathbb{Z}$  is the group of addition of positive and negative integers. As an immediate physical application, we note that when a gauge group  $SO(2)$  is completely spontaneously broken, the coset space is just  $SO(2)$  itself, which has the topology of a circle, so in this case there is an infinite number of types of topologically stable vortex line, characterized by a positive or negative integer  $n$ . For instance, this is the case in a type II superconductor, where as we saw in [Section 21.6](#) the  $U(1)$  of electromagnetic gauge invariance is spontaneously broken to a discrete subgroup  $\mathbb{Z}_2$ .

More generally, all spheres  $S_d$  with  $d > 1$  are simply connected, which means that they have a trivial first homotopy group, a statement conventionally expressed as

$$\pi_1(S_d) = 0 \quad \text{for } d > 1. \quad (23.2.3)$$

Only a few of the more familiar Lie groups are multiply connected:

$$\pi_1(G) = \begin{cases} \mathbb{Z}, & G = U(k), \quad k \geq 1, \\ \mathbb{Z}_2, & G = SO(k), \quad k \geq 3, \\ 0, & G = \text{Spin}(k), \quad k \geq 3, \\ 0, & G = SU(k), \quad k \geq 2, \\ 0, & G = USp(2k), \quad k \geq 1, \\ 0, & G = G_2, F_4, E_6, E_7, E_8. \end{cases} \quad (23.2.4)$$

Here  $\mathbb{Z}_2$  is the group with two elements  $\mathbf{1}$  and  $-\mathbf{1}$ , with group multiplication defined as ordinary multiplication, and  $\text{Spin}(n)$  is the simply connected covering group of  $SO(n)$ . (As we saw in [Section 2.7](#),  $\text{Spin}(3)$  is the same as  $SU(2)$ .) Also, for a direct product of two manifolds  $\mathcal{M}$  and  $\mathcal{M}'$ , the fundamental group is

$$\pi_1(\mathcal{M} \cdot \mathcal{M}') = \pi_1(\mathcal{M}) \cdot \pi_1(\mathcal{M}'). \quad (23.2.5)$$

We can appreciate the physical significance of the group structure of  $\pi_1(\mathcal{M})$  by asking what happens when two distant parallel vortex lines in three dimensions are brought together. When the vortex lines are sufficiently far apart their fields do not interact, so the configuration can be described by specifying the classes  $c'$  and  $c''$  in  $\pi_1(G/H)$  to which each belongs. The class  $c$  to which the whole configuration belongs is determined by the behavior of the fields on a very large circle surrounding both vortex lines. By a continuous deformation we can distort this very large circle into two large circles, one surrounding each vortex line, which intersect at a point midway between them. As we go around this closed curve in two-dimensional space we first trace out a curve in  $G/H$  which consists of one of the closed curves in the class  $c'$ , and then one of the closed curves in the class  $c''$ , just as in the definition of the product of the classes. We conclude then that the whole configuration is in a class  $c = c' \cdot c''$ , and so the two vortex lines can only fuse together to form a vortex line of this class. In particular, they can only annihilate if  $c'' = c'^{-1}$ .

For instance, when  $\pi_1(G/H) = \mathbb{Z}$  (as in superconductivity, where  $G/H = SO(2)/\mathbb{Z}_2$ ), in three dimensions two vortex lines of classes  $c_n$  and  $c_m$  can fuse together to form a vortex line of class  $c_{n+m}$ , so they can only annihilate if  $n = -m$ . On the other hand, when  $\pi_1(G/H) = \mathbb{Z}_2$  (as when  $G = SO(N)$  with  $N \geq 3$  and  $H$  is trivial or discrete) there is only one kind of vortex line in three dimensions, corresponding to the element  $-\mathbf{1}$  of  $\mathbb{Z}_2$ , and since  $(-\mathbf{1})^2 = \mathbf{1}$ , any two can annihilate.

Now let us consider the general homotopy group  $\pi_k(\mathcal{M})$ . This is much like  $\pi_1(\mathcal{M})$ , except that instead of considering mappings of the circle  $S_1$  into a manifold  $\mathcal{M}$ , we consider mappings of the  $k$ -sphere  $S_k$  (the surface of a  $(k+1)$ -dimensional ball) into  $\mathcal{M}$ , again with one point of  $S_k$  always mapped into the same ‘base point’  $p_0$  of  $\mathcal{M}$ . Two such mappings are equivalent if one can be continuously deformed into the other, keeping the same point of  $S_k$  always mapped into the base point. The  $k$ th homotopy group  $\pi_k(\mathcal{M})$  has elements consisting of equivalence classes of these mappings.

It is often convenient to picture the  $d$ -sphere  $S_d$  as the interior of a  $d$ -dimensional hypercube, with all points on the boundary identified as a single point. For instance, we have already seen that the circle  $S_1$  can be treated as the interval  $0 \leq \theta \leq 2\pi$ , with points  $\theta = 0$  and  $\theta = 2\pi$  identified. Similarly we can make a map of an  $S_2$  like the earth’s surface by cutting out the south pole and spreading out the resulting sheet on the unit square  $0 \leq z_1 \leq 1, 0 \leq z_2 \leq 1$ . Continuous mappings of this square into  $\mathcal{M}$  must take all points on the boundary into the same point of  $\mathcal{M}$ , because all points on the boundary are the same point, the south pole. In general, two mappings  $p(z_1, \dots, z_d)$  and  $p'(z_1, \dots, z_d)$  of  $S_d$  into  $\mathcal{M}$  are homotopically equivalent if one can be continuously deformed into the other, while keeping  $p$  on the boundary of the hypercube equal to the base point  $p_0$ .

As before, for each equivalence class  $c$  we choose a standard mapping  $p(z_1, \dots, z_d; c)$ . The product of  $c_1$  and  $c_2$  is defined as the equivalence class that contains the mapping

$$p(z_1, z_2, \dots, z_d; c_1, c_2) = \begin{cases} p(2z_1, z_2, \dots, z_d; c_1), & 0 \leq z_1 \leq \frac{1}{2}, \\ p(2z_1 - 1, z_2, \dots, z_d; c_2), & \frac{1}{2} \leq z_1 \leq 1. \end{cases} \quad (23.2.6)$$

The unit element  $e$  is defined as the equivalence class that contains the mapping with  $p = p_0$  for all  $z$ , and the inverse  $c^{-1}$  of  $c$  is defined as the equivalence class that contains the mapping with

$$p^{-1}[z_1, z_2, \dots, z_N; c] = p[1 - z_1, z_2, \dots, z_N; c]. \quad (23.2.7)$$

In the same way as for  $\pi_1$ , it can be shown that this multiplication is associative, and that  $e \cdot c = c \cdot e = c$  and  $c^{-1} \cdot c = c \cdot c^{-1} = e$ . All  $\pi_n(\mathcal{M})$  for  $n \geq 2$  are Abelian. (There are manifolds  $\mathcal{M}$  for which  $\pi_1(\mathcal{M})$  is non-Abelian, such as the plane with two or more points removed.)

In any case in which  $\pi_k(\mathcal{M}) = \mathbb{Z}$ , there must be a one-to-one mapping of the  $k$ -sphere  $S_k$  into a  $k$ -sphere  $S'_k$  in  $\mathcal{M}$ , which corresponds to the element ‘one’ of  $\mathbb{Z}$  (not the unit element, which is zero). The element  $\nu$  of  $\mathbb{Z}$  with  $\nu = 2, 3, \dots$  corresponds to the mapping of  $S_k$  into the same  $k$ -sphere  $S'_k$  in  $\mathcal{M}$ , which covers  $S'_k$   $\nu$  times, with the Jacobian of the transformation  $S_k \rightarrow S'_k$  positive. The element  $\nu$  of  $\mathbb{Z}$  with  $\nu = -1, -2, \dots$  corresponds to the mapping  $S_k \rightarrow S'_k$ , which covers  $S'_k$   $|\nu|$  times, with the Jacobian of the transformation  $S_k \rightarrow S'_k$  negative.

For instance, we saw in the previous section that magnetic monopoles arise when a simply connected group  $G$  is broken to the  $U(1)$  of electromagnetism. In this case the appendix shows that

$$\pi_2(G/U(1)) = \pi_1(U(1)) = \mathbb{Z}, \quad (23.2.8)$$

so a magnetic monopole carries an integer-valued quantum number  $\nu$ , which as shown in [Section 23.3](#) is proportional to the magnetic charge. This quantum number gives the number of times that a two-sphere of large radius surrounding the monopole is mapped into a two-sphere in the manifold  $G/U(1)$  of Goldstone boson fields (with the relative orientation of the two two-spheres being the same or opposite for  $\nu$  positive or negative, respectively) and is therefore known as the *winding number*. The structure of the group  $\mathbb{Z}$  shows that this quantum number is conserved, in the sense that a monopole of quantum number  $\nu$  can fuse with a monopole of quantum number  $\nu'$  only to form a monopole with quantum number  $\nu + \nu'$ .

If the unbroken subgroup is  $SO(n)$  with  $n \geq 3$ , then according to [Appendix B of this chapter](#)

$$\pi_2(G/SO(n)) = \pi_1(SO(n)) = \mathbb{Z}_2. \quad (23.2.9)$$

In this case there is just one sort of ‘monopole,’ corresponding to the element  $-1$  of  $\mathbb{Z}_2$ , which can annihilate only in pairs. It is important to distinguish between this case and that in which  $SO(n)$  is replaced with its simply connected covering group  $\text{Spin}(n)$ , for which there are no monopoles. We will come back to this point at the end of the next section.

Another example: we saw in the previous section that the skyrmions in quantum chromodynamics with  $n$  light quarks correspond to elements of  $\pi_3(SU(n))$ , which according to the appendix is  $\mathbb{Z}$ . These skyrmions carry a conserved integer-valued quantum number  $\nu$ , which perhaps may be identified with baryon number. Similarly, recall from the previous section that the instantons in a gauge theory based on a simple gauge group  $G$  correspond to the elements of  $\pi_3(G)$ , which according to the appendix is  $\mathbb{Z}$ , so these instantons like skyrmions carry an integer-valued quantum number  $\nu$ , also known as the winding number. In [Section 23.5](#) we shall see how to express this quantum number as a local functional of the gauge field.

### 23.3 Monopoles

As a detailed example of a topologically non-trivial field configuration, we shall now consider the monopole of 't Hooft and Polyakov [\[2\]](#), and its generalizations. We saw in [Section 23.1](#) that when a simply connected gauge group  $G$  is spontaneously broken to the  $U(1)$  of electromagnetism, the configurations of finite energy are classified according to the elements of the group  $\pi_2(G/U(1)) = \pi_1(U(1)) = \mathbb{Z}$ . (The case of non-simply connected Lie groups is considered at the end of this section.) According to the physical interpretation of homotopy groups discussed in [Section 23.2](#), this means that these configurations have a conserved additive quantum number. But we still need to show that any of these stationary configurations actually exist, and to give a physical interpretation of their topological quantum numbers.

As an illustrative example, consider a theory (like the Georgi–Glashow electroweak model [\[12\]](#)) in which an  $SU(2)$  gauge group is spontaneously broken by the vacuum expectation value of an  $SU(2)$  triplet of scalar fields  $\phi_n$ . (It is explained at the end of this section why in this case we say that the gauge group is  $SU(2)$  rather than  $SO(3)$ .) The Lagrangian density for the scalars and gauge fields in Minkowskian spacetime is taken as

$$\mathcal{L} = -\frac{1}{4}F_{n\mu\nu}F_n^{\mu\nu} - \frac{1}{2}D_\mu\phi_n D^\mu\phi_n - V(\phi_n\phi_n), \quad (23.3.1)$$

where

$$F_{n\mu\nu} \equiv \partial_\mu A_{n\nu} - \partial_\nu A_{n\mu} + e\epsilon_{nml}A_{m\mu}A_{l\nu}, \quad (23.3.2)$$

$$D_\mu\phi_n \equiv \partial_\mu\phi_n + e\epsilon_{nml}A_{m\mu}\phi_l, \quad (23.3.3)$$

and the function  $V(\phi_n\phi_n)$  is assumed to be positive, with the value zero at a non-vanishing value  $\langle\phi\rangle$  (taken positive) of  $\sqrt{\phi_n\phi_n}$ . (In much work on monopoles  $V$  is taken to be the quartic polynomial  $\lambda(\phi_n\phi_n - \langle\phi\rangle^2)^2$ , with  $\lambda > 0$ , but we shall not make this assumption here.) Eq. [\(23.3.1\)](#) describes a theory with a spacetime-independent vacuum solution with  $A_{n\mu} = 0$ , in which a vacuum expectation value  $\phi_n$  with  $\phi_n\phi_n = \langle\phi\rangle^2$  breaks  $SU(2)$  to its  $U(1)$  subgroup, which can be identified with the gauge group of electrodynamics. Instead we here shall seek topologically non-trivial inhomogeneous but time-independent classical solutions in temporal gauge, with  $A_n^0 = 0$  but  $A_n^i \neq 0$ . The Lagrangian density in this case is the negative of the potential energy density  $\mathcal{H}$ , given by

$$\mathcal{H} = \frac{1}{4}F_{nij}^2 + \frac{1}{2}(D_i\phi_n)^2 + V(\phi), \quad (23.3.4)$$

with squares including obvious index contractions. Because each term in (23.3.4) is positive, the integral of each term must separately converge in a configuration of finite energy.

In particular, for the integral of  $V(\phi_n\phi_n)$  to converge, the vector  $\phi_n$  must have the fixed length  $\langle\phi\rangle$  at infinity, so each configuration of finite energy defines a smooth mapping of a large two-sphere  $\mathcal{S}$  surrounding the monopole configuration into the two-sphere of the  $\phi_n$  with  $\phi_n\phi_n = \langle\phi\rangle^2$ . As  $\hat{\mathbf{x}}$  runs over  $\mathcal{S}$ ,  $\phi_n$  may run over the sphere  $\phi_n\phi_n = \langle\phi\rangle^2$  any integer number  $N$  of times, either with Jacobian  $\text{Det}(\partial\mathbf{x}/\partial\phi)$  positive, in which case we say that the winding number is  $N$ , or with Jacobian negative, in which case the winding number is  $-N$ .

To see what the winding number has to do with the magnetic monopole moment, we must first consider what in this theory is observed as ‘the’ magnetic field. Whatever the field configuration, we can introduce a gauge in which the scalar field  $\phi_n$  points in some definite direction, say the three-direction, in any given finite region, so that in this region the gauge field associated with the unbroken  $U(1)$  subgroup of  $SU(2)$  is  $A_{3i}$ . ’t Hooft [2] found a gauge-invariant tensor which reduces to the usual electromagnetic field strength tensor  $\partial_\mu A_{3\nu} - \partial_\nu A_{3\mu}$  in this gauge:

$$\mathcal{F}_{\mu\nu} \equiv F_{n\mu\nu}\hat{\phi}_n - \frac{1}{e}\epsilon_{nml}\hat{\phi}_n D_\mu\hat{\phi}_m D_\nu\hat{\phi}_l, \quad (23.3.5)$$

where  $\hat{\phi}_n \equiv \phi_n/\sqrt{\phi_m\phi_m}$ . To check that  $\mathcal{F}_{\mu\nu}$  is the ordinary electromagnetic field strength tensor in a gauge with constant  $\hat{\phi}_n$  (and for later purposes) we use Eqs. (23.3.2) and (23.3.3) and the identity  $\epsilon_{abc}\epsilon_{ade} = \delta_{bd}\delta_{ce} - \delta_{be}\delta_{cd}$  to write  $\mathcal{F}_{\mu\nu}$  in the form [13]

$$\mathcal{F}_{\mu\nu} = \partial_\mu(\hat{\phi}_n A_{n\nu}) - \partial_\nu(\hat{\phi}_n A_{n\mu}) - \frac{1}{e}\epsilon_{nml}\hat{\phi}_n \partial_\mu\hat{\phi}_m \partial_\nu\hat{\phi}_l. \quad (23.3.6)$$

Thus in a gauge in which  $\hat{\phi}_n$  is a fixed unit vector in the three-direction, we have as promised

$$\mathcal{F}_{\mu\nu} = \partial_\mu A_{3\nu} - \partial_\nu A_{3\mu}.$$

The magnetic monopole moment  $g$  of any localized field configuration is defined as  $1/4\pi$  times the magnetic flux through a large closed surface  $\mathcal{S}$  around the configuration:

$$4\pi g \equiv \frac{1}{2}\epsilon_{ijk} \int_{\mathcal{S}} \mathcal{F}_{ij} d^2 S_k. \quad (23.3.7)$$

The first two terms in Eq. (23.3.6) for  $\mathcal{F}_{ij}$  are derivatives, and therefore do not contribute to the integral (23.3.7), so that

$$g = -\frac{1}{8\pi e}\epsilon_{ijk}\epsilon_{nml} \int_{\mathcal{S}} \hat{\phi}_n \partial_i \hat{\phi}_m \partial_j \hat{\phi}_l d^2 S_k. \quad (23.3.8)$$

This has the important property of being a topological invariant: integrating by parts where necessary, we can see that the change in  $g$  under an infinitesimal variation  $\delta\hat{\phi}_n$  in  $\hat{\phi}_n$  is

$$\delta g = -\frac{3}{8\pi e}\epsilon_{ijk}\epsilon_{nml} \int_{\mathcal{S}} \delta\hat{\phi}_n \partial_i \hat{\phi}_m \partial_j \hat{\phi}_l d^2 S_k.$$

But because  $\hat{\phi}$  is a unit vector,  $\delta\hat{\phi}$  as well as  $\partial_i\hat{\phi}$  and  $\partial_j\hat{\phi}$  are all in the plane perpendicular to  $\hat{\phi}$ , so

$$\epsilon_{nml}\delta\hat{\phi}_n\partial_i\hat{\phi}_m\partial_j\hat{\phi}_l = 0,$$

and therefore  $\delta g = 0$ . The quantity (23.3.8) is related to the topological invariant known as the *Kronecker index*.

Because  $g$  is a surface integral, it is additive: for any two distant localized configurations, the surface  $\mathcal{S}$  used to calculate  $g$  may be taken to be a pair of spheres, one surrounding each configuration, connected by a thin neck between them, so the value of  $g$  for the whole system will be the sum of the values of  $g_1$ ,  $g_2$  for the individual localized configurations. Furthermore, since  $g$  is a topological invariant, we will have  $g = g_1 + g_2$  for any field configuration that is formed by a smooth fusion of two configurations with magnetic monopole moments  $g_1$  and  $g_2$ . It follows that  $g$  must be proportional to the winding number. Arafune, Freund, and Goebel [13] verified this and calculated the coefficient of proportionality by using the formula (23.3.8) for general winding number. Here we will simply calculate the coefficient by studying the 't Hooft–Polyakov monopole [2], in which the fields have unit winding number.

As we saw in the previous section, the ‘identity’ (as opposed to the unit) element  $\mathcal{I}$  of  $\pi_2(SU(2)/U(1))$ , which corresponds to the element ‘one’ of  $\mathbb{Z}$ , consists of configurations in which the two-sphere  $\mathcal{S}$  at infinity is mapped once (with positive Jacobian) into the sphere described by  $\hat{\phi}_n$ . As a representative of this homotopy class, we may take a configuration in which at infinity  $\phi_n$  is in the same direction as  $\mathbf{x}$ . To construct this configuration let us impose a symmetry under joint rotations on  $\phi = \{\phi_1, \phi_2, \phi_3\}$  and  $\mathbf{x}$ , as well as parity conservation, and make the ansatz:

$$\phi_n = \hat{x}_n \langle \phi \rangle F(r), \quad (23.3.9)$$

$$A_{ni} = \frac{\epsilon_{nil}\hat{x}_l}{er} G(r). \quad (23.3.10)$$

There is an important similarity between this field configuration and that for a vortex line in a superconductor. The solution  $\phi = \pm \ell \varphi / 2e$  for the Goldstone boson field in a vortex line found in Section 21.6 shows that, although gauge invariance and rotational invariance are spontaneously broken, the vortex line solution is invariant under the combination of a global gauge transformation for which  $\phi \rightarrow \phi + \Lambda$  and a rigid rotation  $\varphi \rightarrow \varphi \pm 2e\Lambda/\ell$ . Similarly, a monopole solution of the form (23.3.9)–(23.3.10) is not invariant under rotations or gauge transformations, but it is invariant under the combination of a rigid three-dimensional spatial rotation and an equal global  $SO(3)$  gauge transformation.

As already mentioned, in order for the integral of  $V(\phi_n\phi_n)$  to converge it is necessary for  $\phi_n\phi_n$  to approach  $\langle \phi \rangle^2$  as  $r \rightarrow \infty$ , so in this limit  $F(r) \rightarrow 1$ . To derive the limiting behavior of  $G(r)$ , note that the covariant derivative of the scalar field is

$$D_i\phi_n = \pm \langle \phi \rangle \left[ (1 - G(r))(\delta_{ni} - \hat{x}_n\hat{x}_i) \frac{F(r)}{r} + \hat{x}_n\hat{x}_i F'(r) \right], \quad (23.3.11)$$

so the scalar term in the Hamiltonian density is

$$\frac{1}{2}(D_i\phi_n)^2 = \langle \phi \rangle^2 \left[ \frac{F^2(1 - G)^2}{r^2} + \frac{F'^2}{2} \right]. \quad (23.3.12)$$



For this to have a finite integral it is necessary also that  $G(r) \rightarrow 1$  and  $F'(r) \rightarrow 0$  as  $r \rightarrow \infty$ . Finally, the field strength (23.3.2) is

$$F_{nij} = \frac{\epsilon_{ijk}}{e} \left[ -\frac{1}{r} G'(r) (\delta_{nk} - \hat{x}_n \hat{x}_k) - \frac{1}{r^2} (2G(r) - G^2(r)) \hat{x}_n \hat{x}_k \right], \quad (23.3.13)$$

so the Yang–Mills term in the Hamiltonian density is

$$\frac{1}{4} (F_{nij})^2 = \frac{1}{e^2} \left[ \frac{G'^2}{r^2} + \frac{(2G - G^2)^2}{2r^4} \right]. \quad (23.3.14)$$

This has an integral that converges at large distances as long as  $G'(r)$  vanishes sufficiently rapidly as  $r \rightarrow \infty$ .

We can use these results to calculate the magnetic monopole moment of this configuration. Eq. (23.3.13), together with the limits  $G(\infty) = 1$  and  $G'(\infty) = 0$ , shows that for  $r \rightarrow \infty$ , we have

$$F_{nij} \longrightarrow -\frac{\epsilon_{ijk} \hat{x}_k \hat{x}_n}{er^2}. \quad (23.3.15)$$

Since  $D_i \phi_n$  vanishes rapidly as  $r \rightarrow \infty$ , the magnetic part of the field strength tensor in this gauge is given for  $r \rightarrow \infty$  by the first term in Eq. (23.3.5), so at large distances the magnetic field becomes

$$B_i \equiv \frac{1}{2} \epsilon_{ijk} \mathcal{F}_{jk} \longrightarrow \frac{1}{2} \epsilon_{ijk} \hat{\phi}_n F_{njk} \longrightarrow \frac{\hat{x}_i}{er^2}. \quad (23.3.16)$$

Thus *this configuration has magnetic monopole moment  $g = 1/e$* . According to the general argument above, the magnetic moment of a configuration with winding number  $\nu$ , corresponding to the element  $\nu$  of  $\mathbb{Z}$ , is then

$$g_\nu = \nu/e. \quad (23.3.17)$$

It requires a detailed numerical calculation to find the stable configuration that minimizes the integral of the Hamiltonian density (23.3.4). However there is a limiting case where an analytic solution is available. To see this, it is useful first to derive a general lower bound due to Bogomol'nyi [10] on the monopole energy for a given magnetic monopole moment  $g$ . Note that (23.3.4) may be written

$$\mathcal{H} = \frac{1}{4} (F_{nij} \mp \epsilon_{ijk} D_k \phi_n)^2 \pm \frac{1}{2} \epsilon_{ijk} F_{nij} D_k \phi_n + V(\phi_n \phi_n). \quad (23.3.18)$$

Using the Bianchi identity (15.3.9), the second term may be written

$$\pm \frac{1}{2} \epsilon_{ijk} F_{nij} D_k \phi_n = \pm \frac{1}{2} \epsilon_{ijk} D_k (F_{nij} \phi_n) = \pm \frac{1}{2} \epsilon_{ijk} \partial_k (F_{nij} \phi_n),$$

so its integral is given by the magnetic monopole moment  $g$

$$\pm \frac{1}{2} \epsilon_{ijk} \int d^3x F_{nij} D_k \phi_n = \pm \langle \phi \rangle \int \mathbf{B} \cdot d\mathbf{A} = \pm 4\pi \langle \phi \rangle g.$$

Since every other term in  $\mathcal{H}$  is positive, we have a general lower bound on the energy of a configuration with magnetic monopole moment  $g$

$$E = \int d^3x \mathcal{H} \geq 4\pi \langle \phi \rangle |g|. \quad (23.3.19)$$

For  $g = \pm 1/e$ , this gives an energy  $E \geq 4\pi \langle \phi \rangle / e$ , which for small coupling constant  $e$  is much greater than the corrections due to quantum fluctuations, which are at most of order  $\langle \phi \rangle$ . This is why we can take such a classical configuration seriously as the leading term in a perturbation expansion.

Now, it is tempting to try to minimize the energy for a given magnetic monopole moment by setting the first term in Eq. (23.3.18) equal to zero, so that

$$F_{nij} = \pm \epsilon_{ijk} D_k \phi_n, \quad (23.3.20)$$

but in general this does not lead to a configuration at which the energy is stationary. The condition that the energy should be stationary with respect to variations in the scalar field is the field equation

$$D_k D_k \phi_n = 2\phi_n V'(\phi_n \phi_n), \quad (23.3.21)$$

while Eq. (23.3.20) together with the Bianchi identity (15.3.9) would imply that  $D_k D_k \phi_n = 0$ . This argument suggests that in the special case where  $V(\phi_n \phi_n)$  is very small, it will be possible nearly to reach the lower bound (23.3.19), and hence minimize the energy for a given magnetic monopole moment, by imposing the condition (23.3.20). (Where  $V$  is the quartic polynomial  $\lambda(\phi_n \phi_n - \langle \phi \rangle^2)^2$ , the assumption that  $V$  is small means that  $\lambda \ll e^2$ , as in a type I superconductor.) Such stable configurations were studied in this way by Bogomol'nyi [10]. They had been found earlier by Prasad and Sommerfield [14] without direct use of Eq. (23.3.20), and are usually called *BPS monopoles*.

The Bogomol'nyi condition (23.3.20) provides first-order differential equations for  $F(r)$  and  $G(r)$ , which are much easier to solve than the second-order field equations derived directly from the condition that the energy should be stationary. Using Eqs. (23.3.11) and (23.3.13), the terms in Eq. (23.3.20) proportional to  $\epsilon_{ijk}[\delta_{kn} - \hat{x}_k \hat{x}_n]$  and  $\epsilon_{ijk} \hat{x}_k \hat{x}_n$  respectively yield the differential equations

$$e \langle \phi \rangle F(1 - G) = G', \quad (23.3.22)$$

$$e \langle \phi \rangle r^2 F' = G(2 - G). \quad (23.3.23)$$

With the boundary condition that  $F(r) \rightarrow 1$  and  $G(r) \rightarrow 1$  as  $r \rightarrow \infty$ , these equations have the solution

$$F = \coth \rho - \frac{1}{\rho}, \quad G = 1 - \frac{\rho}{\sinh \rho}, \quad (23.3.24)$$

where  $\rho = e \langle \phi \rangle r$ . Note in particular that the field  $\phi_n$  given by Eqs. (23.3.9) and (23.3.24) vanishes for  $r \rightarrow 0$ , so as remarked in Section 23.1, the  $SU(2)$  symmetry is restored at the center of the monopole.

Let's now return to the case of a potential  $V$  of arbitrary strength. The 't Hooft–Polyakov monopole is stable, because there are no configurations of smaller topological quantum

number into which it could decay. Configurations of higher magnetic monopole moment are generally unstable [10]. There are also interesting configurations with both magnetic monopole moments and electric charge, known as *dyons* [15].

There is another way of understanding the value  $1/e$  of the 't Hooft–Polyakov magnetic monopole moment, which goes back to the original work of Dirac on magnetic monopoles [16]. As mentioned earlier, instead of the gauge we have been using, we can make a gauge transformation  $\phi_n \rightarrow R_{nm}(\hat{\mathbf{x}})\phi_m$  that rotates  $\phi_n$  to point in a fixed direction  $\hat{\mathbf{v}}$ , for instance the three-direction. Then the field strength  $B_{nk} \equiv \frac{1}{2}\epsilon_{ijk}F_{nij}$  is transformed into  $R_{nm}B_{mk}$ , which approaches  $\hat{v}_n\hat{x}_k/er^2$  for  $r \rightarrow \infty$ , so here we do not have to project this on a local unbroken symmetry direction. The price we need to pay for these conveniences is that the gauge transformation is singular; the rotation that takes a vector in the  $\hat{\mathbf{x}}$  direction into some fixed direction  $\hat{\mathbf{v}}$  is

$$\begin{aligned} R(\hat{\mathbf{x}}; \hat{\mathbf{v}}) = & \mathbf{1} - (1 - \hat{\mathbf{v}} \cdot \hat{\mathbf{x}})\hat{\mathbf{v}}\hat{\mathbf{v}}^T + \hat{\mathbf{v}}(\hat{\mathbf{x}} - (\hat{\mathbf{x}} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}})^T \\ & + (\hat{\mathbf{x}} - (\hat{\mathbf{x}} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}})\hat{\mathbf{v}}^T \\ & - \frac{(\hat{\mathbf{x}} - (\hat{\mathbf{x}} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}})(\hat{\mathbf{x}} - (\hat{\mathbf{x}} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}})^T}{1 + \hat{\mathbf{x}} \cdot \hat{\mathbf{v}}}, \end{aligned} \quad (23.3.25)$$

which is singular at  $\hat{\mathbf{x}} = -\hat{\mathbf{v}}$ . This  $R$  is not unique; for instance, we could perform the rotation  $R(\hat{\mathbf{x}}; -\hat{\mathbf{v}})$  that takes  $\hat{\mathbf{x}}$  into the  $-\hat{\mathbf{v}}$  direction, followed by a fixed rotation of  $180^\circ$  around some axis perpendicular to  $\hat{\mathbf{v}}$ , but this would become singular at  $\hat{\mathbf{x}} = +\hat{\mathbf{v}}$ . To avoid the singularities, we have to adopt different gauges in different regions; for instance, with  $\hat{\mathbf{v}}$  in the three-direction, we can use a gauge for  $0 < \theta < \theta_0$  that is singular at  $\theta = \pi$  and a gauge for  $\theta_0 < \theta < \pi$  that is singular at  $\theta = 0$ , where  $\theta_0$  is an arbitrary angle with  $0 < \theta_0 < \pi$ , often taken as  $\pi/2$ . Everywhere except at  $\theta = 0$  and  $\theta = \pi$  the magnetic field will be given at large distances by  $\mathbf{B} \rightarrow g\hat{\mathbf{x}}/r^2$ , where  $g$  is the magnetic monopole strength. This can be written as the curl of a vector potential whose only non-vanishing component is in the azimuthal  $\varphi$  direction. For  $0 < \theta < \theta_0$  we must take  $A_\varphi = g(1 - \cos\theta)/r \sin\theta$ , which is only singular at  $\theta = \pi$ , while for  $\theta_0 < \theta < \pi$  we must take  $A_\varphi = -g(1 + \cos\theta)/r \sin\theta$ , which is only singular at  $\theta = 0$ . The difference  $\Delta\mathbf{A}$  between these two vector potentials is a gradient  $\nabla\Lambda$  with  $\Lambda = 2g\varphi$ , which of course does not affect the magnetic field for  $0 < \theta < \pi$ , but could affect the dynamics of charged fields. A gauge transformation with  $\Lambda = 2g\varphi$  will change a field of charge  $q$  by a factor  $\exp(2iqg\varphi)$ , which is not single-valued unless  $2qg$  is an integer. This is the Dirac quantization condition; the existence of any magnetic monopole with monopole moment  $g$  would require all electric charges to be integer multiples of  $(2g)^{-1}$ . For the 't Hooft–Polyakov monopole this condition is automatically satisfied, because here  $g = 1/e$ , and all charges in the Georgi–Glashow model are integer multiples of  $e/2$ .

The Georgi–Glashow model was ruled out as a theory of weak and electromagnetic interactions by the discovery of neutral currents, but magnetic monopoles are expected to occur in other theories, where a simply connected group  $G$  is spontaneously broken not to  $U(1)$ , but to some subgroup  $H' \cdot U(1)$ , where  $H'$  is simply connected. (According to [Appendix B of this chapter](#), for simply connected groups  $G$  we have  $\pi_2(G/H) = \pi_1(H)$ , which for  $H = H' \cdot U(1)$  equals  $\pi_1(H') \cdot \pi_1(U(1)) = \pi_1(U(1)) = \mathbb{Z}$ .) There are no monopoles

produced in the spontaneous breaking of the gauge group  $SU(2) \cdot U(1)$  of the standard electroweak theory, which is not simply connected. (About this, more below.) But we do find monopoles when the simply connected gauge group  $G$  of theories of unified strong and electroweak interactions, such as  $SU(4) \cdot SU(4)$  or  $SU(5)$  or  $\text{Spin}(10)$ , is spontaneously broken to the gauge group  $SU(3) \cdot SU(2) \cdot U(1)$  of the standard model. (See [Section 21.5](#).) The monopoles in this case are expected to have a mass larger by an inverse square gauge coupling constant than the vector boson masses  $M \approx 10^{15}\text{--}10^{16}$  GeV produced by this symmetry breaking. Such monopoles would have been produced when the universe underwent a phase transition in which  $G$  was spontaneously broken to  $SU(3) \cdot SU(2) \cdot U(1)$ , at a temperature  $T$  of order  $M$ .

This poses a problem for some cosmological models [\[17\]](#). The scalar fields before this phase transition would have necessarily been uncorrelated at distances larger than the horizon distance, the furthest distance that light could have travelled since the initial singularity. At an early time  $t$  in standard cosmological theories [\[18\]](#) the horizon distance is of order  $t \approx (G_N T^4)^{-1/2}$  (where  $G_N \approx (10^{19} \text{ GeV})^{-2}$  is Newton's constant), so the number density of monopoles produced at this time would have been of order  $t^{-3} \approx (G_N M^4)^{3/2}$ , which is smaller than the photon density  $M^3$  at  $T \approx M$  by a factor of order  $(G_N M^2)^{3/2}$ . For  $M \approx 10^{15}$  GeV this factor is of order  $10^{-6}$ . If monopoles did not find each other to annihilate, then this ratio would remain roughly constant to the present, but with at least  $10^9$  microwave background photons per nucleon today, this would give at least  $10^3$  monopoles per nucleon, in gross disagreement with what is observed. This potential paradox was one of the factors leading to inflationary cosmological models [\[19\]](#), in which there was a period of exponential expansion, which if it occurred before the monopoles were produced would have greatly extended the horizon, and if it occurred after the production of monopoles (but before a period of reheating) would have greatly diluted the monopole density.

The discovery of monopoles of any sort would create opportunities for the observation of remarkable phenomena, including the existence of fermion-monopole configurations of fractional fermion number [\[20\]](#), and the violation of baryon conservation in fermion-monopole scattering [\[21\]](#).

\* \* \*

In the above discussion we have considered only monopoles associated with the spontaneous breakdown of a simply-connected gauge group  $G$ . This raises a question. For every Lie group  $G$ , whether simply connected or not, there is a simply-connected group  $\overline{G}$  with the same Lie algebra, known as its *covering group*. (For examples, see [Section 2.7](#).) Any non-simply connected group has fewer representations than its covering group (for instance, the doubly-connected groups  $SO(n)$  have only scalar, vector, and tensor representations, while their covering groups  $\text{Spin}(n)$  also have spinor representations). If a theory does not happen to involve fields that belong to the extra representations of the covering group, then are we free to consider the gauge group of the theory to be either the non-simply connected group  $G$  or its covering group  $\overline{G}$ ? In particular, does the menu of possible monopoles depend on whether the theory contains only fields transforming as representations of a non-simply connected group  $G$ , or additional fields that would only furnish representations of its

covering group  $\overline{G}$ ? For instance, in the original Georgi–Glashow model [12] the only scalar fields belonged to a representation of  $SO(3)$ , the three-vector, but there were fermions that belonged to spinor representations of the covering group  $\text{Spin}(3) = SU(2)$ . Would the allowed values of the magnetic monopole moment change if we added scalar fields belonging to the spinor representations of  $SU(2)$ ? Would it change if we removed the fermions?

The answer is that the menu of possible monopoles does not depend on whether we say that the gauge group is a non-simply connected group  $G$  or its covering group  $\overline{G}$ , and therefore is unaffected when we add or remove fields belonging to representations of  $\overline{G}$  that are not representations of  $G$ . As we saw in Section 23.1, in general the topologically stable monopole-like configurations are classified according to the elements of  $\pi_2(G/H)$ . According to a result quoted in Appendix B of this chapter, this homotopy group consists of those elements of  $\pi_1(H)$  that correspond to the trivial element of  $\pi_1(G)$  when  $H$  is embedded in  $G$ . But if we replace  $G$  with its covering group  $\overline{G}$ , we also replace  $H$  with a different subgroup  $H'$ , because some of the loops in  $H$  do not return to the base point when  $H$  is embedded in  $\overline{G}$ . These are just the loops that do not become trivial when  $H$  is embedded in  $G$ , so  $\pi_2(G/H) = \pi_1(H')$ , and thus as far as monopoles are concerned we could just as well say that the gauge group is  $\overline{G}$  rather than  $G$ .

For instance, as far as the scalar fields are concerned, the gauge group of the Georgi–Glashow model might be considered to be the doubly-connected  $SO(3)$  rather than its simply-connected covering group  $SU(2)$ . The unbroken subgroup is then  $SO(2)$ , in which we identify transformations that differ by a  $360^\circ$  rotation. Thus  $\pi_1(SO(2))$  includes loops that extend from the unit element to a  $360^\circ$  rotation, which would not be loops when  $SO(3)$  is embedded in  $SU(2)$ . But  $\pi_2(SO(3)/SO(2))$  is not the same as  $\pi_1(SO(2))$ , but rather excludes the loops that are homotopically non-trivial when  $SO(3)$  is embedded in  $SU(2)$ , which are just the loops that extend from the unit element to a  $360^\circ$  rotation, so  $\pi_2(SO(3)/SO(2))$  is the  $U(1)$  subgroup of  $SU(2)$ , just as if the gauge group were taken to be  $SU(2)$  from the beginning.

It is convenient always to consider the gauge group  $G$  associated with any semi-simple gauge algebra to be the simply-connected covering group, so that we can use the simple result that  $\pi_2(G/H) = \pi_1(H)$ . As we have just seen, the connectivity properties of  $H$  are then fixed by its embedding in  $G$ , or more precisely, by the embedding of the Lie algebra of  $H$  in the Lie algebra of  $G$ . For instance, an  $SU(3)$  gauge algebra might be spontaneously broken either into an  $SU(2)$  subalgebra, under which the defining representation of  $SU(3)$  transforms as a doublet plus a singlet, or into an  $SO(3)$  subalgebra, under which the defining representation of  $SU(3)$  transforms as a three-vector. In the first case we do not have the option of considering the unbroken subgroup to be  $SO(3)$ ; since  $\pi_1(SU(2)) = 0$ , there are no monopoles here. In the second case the unbroken gauge group must be regarded as  $SO(3)$ , not  $SU(2)$ , so the theory does have monopole-like configurations, classified according to the elements of  $\pi_1(SO(3)) = \mathbb{Z}_2$ . The nature of the unbroken subalgebra of  $H$  and its embedding in the gauge algebra  $G$  may be dynamically affected by the variety of field types that we introduce in the Lagrangian, but once the algebra of  $H$  and its embedding in the algebra of  $G$  are fixed, the menu of monopoles is otherwise entirely unaffected by the variety of fields in the theory.

In particular, the argument that led to the Dirac quantization condition shows that, in any theory in which a Lie algebra  $G$  is spontaneously broken to a subalgebra including the electric charge operator, the allowed magnetic monopole moments are integer multiples of the reciprocal of the smallest electric charge that appears in the representations of the covering group of  $G$ , whether or not there is actually any particle that carries that charge in the theory. If the algebra of  $G$  itself contains a  $U(1)$  generator, then we must consider the covering group of this  $U(1)$ , which is the non-compact group of translations along the real line. If this  $U(1)$  generator appears as a term in the electric charge operator, as in the standard electroweak theory, then there is no minimum electric charge in the representations of the covering group, and hence no monopoles.